07 — Paper Track and Small-Model Bonus

These are the two hedges that do not require beating the main 31B leaderboard. Both have lower competition density than the main track, but neither is free: the paper requires a real research contribution, and the small-model bonus requires a working E2B/E4B agent plus one of your two final submission slots.

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1. Paper track

1.1 Facts

- Separate Kaggle competition (S46).
- Total prizes: **$35,000**:
  - Overall best paper: **$15,000**.
  - Best new resource (dataset, library, tool): **$10,000**.
  - Best new application: **$10,000**.
- Deadline: **2026-11-12, 11:59 PM UTC** (S46).
- Scope: **3,000 words maximum**, original, unpublished research (S46).
- Non-archival: submitting does not prevent submitting to other venues (S46).
- Submission: a Kaggle Writeup with title/subtitle, abstract, introduction, methods/experiments, related works/citations. Optional public notebook and/or arXiv-ready PDF (S46).
- Judging: novelty, quality, relevance, verifiability, clarity; each 0–5; final score is the average (S46).
- At retrieval: 5,620 entrants, 338 participants, 327 teams, 348 submissions (S46).

1.2 What the judges are asking for

Criterion	What it means	What to show
Novelty	New insights, deeper understanding, important properties	A clear claim that is not obvious from the public notebooks

Criterion	What it means	What to show
Quality	General and universal beyond the competition	Method that transfers to other SWE agents/models
Relevance	Impact on software engineering/agentic learning	Why the result matters beyond the leaderboard
Verifiability	Enough information to understand and reproduce	Data, code, prompts, configs, hashes, negative results
Clarity	Clearly presented and written	Tight structure, figures/tables, no hype

The paper track is not a “leaderboard diary.” A case study can win if it is rigorous and generalizable. A new dataset/tool/application can win the resource/application prizes even if the main score is modest.

1.3 Strong paper angles from this competition

These are grounded in public evidence and your own experiments:

1. **Evaluator drift and score noise.** The scorer changed repeatedly (swegemma 0.2.7 → 0.2.11, adk-submission 0.2.11 → 0.2.13, vLLM rebuilds) and identical packages scored differently (S39). A rigorous case study on how to compare agents when the evaluator moves is novel and useful. The public Part 2 blog is a model (S39).
2. **Trajectory quality vs quantity for LoRA.** Public evidence: 697 per-call samples did not beat the base; 1,500 trajectories were generated; quality filtering can make a 10% subset outperform the full set (S19, S25). A controlled study on SWE-agent trajectories, with step/segment masking, is relevant.
3. **Budget-aware agent design.** The public prompt-only config’s (author-reported 0.24) edit-by-call-12 rule and 28-call/8-min budget is a concrete intervention (S36–S38). A study of “when to stop exploring and edit” across models/tasks is generalizable.
4. **Local-to-hidden transfer.** Public teams’ local screens did not predict hidden scores (S35, S39). A methodological paper on repository-grouped holdouts, paired comparisons, and noise floors is valuable.
5. **Graph tools for code navigation.** The competition provides AST graphs/embeddings; public audits found them weak (S36–S38), and async functions are missing (S33). A careful study of when graph tools help (symbol

resolution, callers, cross-file dependencies) is a “code comprehension” contribution.

6. **Small-model distillation.** E4B/E2B with teacher trajectories from 31B/stronger models; the small-model bonus provides a natural evaluation. A resource paper with a public trajectory dataset and a reproducible training recipe fits the “new resource” prize.
7. **Test-time verification.** Agents can’t see hidden tests; how much self-verification helps vs costs time? A controlled study of verification policies (one test, two tests, inline assertions) is relevant.

Avoid:

- “We got 0.24 with prompt X” without a generalizable insight.
- A pure leaderboard report with no controls.
- Claims that local gains transfer without paired evidence.
- Hiding negative results; the paper track values verifiability.

1.4 Suggested paper outline (3,000 words)

1. **Title/subtitle** — specific claim, not “Gemma 4 Agent.”
2. **Abstract** — problem, method, key result, implication (150–200 words).
3. **Introduction** — why offline SWE agents matter; what is missing; your contribution (400–500 words).
4. **Related work** — SWE-bench/SWE-agent, LoRA/QLoRA for code agents, trajectory distillation, budgeted agents (400–500 words).
5. **Method** — agent scaffold, data generation, filters, training, evaluation; enough to reproduce (700–900 words).
6. **Experiments** — datasets/splits, metrics, paired design, results, ablations, negative results (700–900 words).
7. **Discussion/limitations** — noise, evaluator drift, transfer, compute (300–400 words).
8. **Conclusion** — one paragraph.
9. **References** — cited works.
10. **Appendix/links** — code, data, prompts, configs, hashes (not counted in the 3,000 words if linked).

Figures/tables:

- A pipeline diagram.

- A paired comparison table with discordant counts.
- A budget/time vs resolve plot.
- A data-quality ablation.
- A failure taxonomy.

1.5 Verifiability package

Prepare:

- Public code repository (Apache 2.0 or compatible).
- Public trajectory dataset (if allowed by licenses/rules).
- Exact prompts/configs/eval settings.
- Model/adapter revisions and hashes.
- Run manifests and per-task results.
- A README that reproduces the main table with commands.
- A limitations section that states what did not work.

Winners must provide training code and methodology (S3). Build the package as you go; don't reconstruct it after winning.

1.6 Timeline

- **Now–Nov 1:** choose angle; collect data/results; write outline.
- **Nov 1–7:** draft method/experiments; make figures; run missing controls.
- **Nov 7–10:** internal review for novelty/verifiability/clarity; check citations.
- **Nov 10–11:** finalize 3,000 words; publish code/data; submit.
- **Nov 12:** deadline.

Do not wait for the main leaderboard result. A paper on a well-controlled negative result can be stronger than a thin "we placed" report.

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2. Small-model bonus

2.1 Facts

- **$10,000** for the highest-ranking submission using Gemma **E2B or E4B** (S47).

- Must explicitly use E2B/E4B as the base model (S47).
- Must be one of the team's **two selected final submissions** (S47).
- LoRA fine-tuning is allowed; same submission rules/tie-breaking as the main track (S47).
- Cross-track eligible: small-model entries remain eligible for main leaderboard awards and the Paper Track (S47).
- A separate daily notebook displays filtered small-model standings (S47).
- [COMMUNITY] Best small-model public score ~**0.06** as of 2026-10-20 (S47; one snapshot, not a measured floor).

2.2 Why it is attractive

- The main track top is 0.24 with 3,548 teams; the small-model bar is ~0.06 in the community snapshot (S4, S47).
- E4B LoRA trains on ~17 GB (S10), so one A100 40 GB is enough; iteration is cheap.
- The bonus is additive and cross-eligible; a small-model entry can still win main/paper awards.

2.3 Why it is risky

- E4B solved **0/31** tasks in one public comparison, while 31B solved 9–11/31 (S34). It submits a patch on ~97% of tasks (S34); the "often edits the wrong file" part is an inference from its 0/31 solve rate, not file-level data [INFERENCE].
- The small-model lane still requires a working agent; a weak scaffold will score 0.
- **Selecting** it consumes one of your two final slots; a probe does not. If your 31B entries are strong, the opportunity cost of selection may exceed $10k expected value.

2.4 Recommended approach

1. Reproduce the 31B prompt-only baseline first; don't start with E4B.
2. Build a separate E4B agent with the same prompt/budget discipline; adjust for its weaker localization:
    - more emphasis on reading the issue's exact names/messages;
    - earlier edit (call 8–10);
    - fewer tools/calls (E4B is fast but wanders);

    - targeted verification.
3. Train E4B LoRA on E4B or 31B trajectories:
    - E4B LoRA ~17 GB; one A100 40 GB.
    - rank 8–16; alpha = rank; text-only; freeze vision/audio.
    - Use the same thinking mode as inference.
4. Validate on a small-model holdout; compare to the E4B prompt-only baseline.
5. Submit as a final candidate only if it has a real score and does not displace a stronger 31B entry.

2.5 Expected-value framing

Let:

- `P_win` = probability your E4B entry is the highest eligible small-model submission.
- `V` = $10,000.
- `C` = opportunity cost of the final slot (expected value of the 31B entry you would otherwise select).

Select E4B only if `P_win × V > C`. With a best score of 0.06 and many teams likely trying, `P_win` may be low unless your E4B agent is clearly better than the public baseline. If your 31B entry has a realistic chance at a main prize, the slot is probably worth more there.

The bonus is most attractive if:

- your 31B post-training line fails;
- you have a strong E4B-specific scaffold/data recipe;
- you have a spare final slot after securing a strong 31B score.

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3. Combining the tracks

A sensible portfolio:

Slot/asset	Use
Final submission 1	Best validated 31B bundle (prompt-only or adapter)
Final submission 2	Diverse 31B challenger **or** E4B bonus entry
Paper track	Independent; submit a rigorous case study/resource

Slot/asset	Use
Public code/data	Supports paper/winner obligations and community

Do not put all effort into the paper and neglect the main submission; the paper deadline is earlier (Nov 12), so front-load the writing if you pursue it. Do not put all effort into E4B and leave the 31B baseline unscored.

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4. Paper/small-model checklist

Paper

- ☐ Angle chosen; novelty claim in one sentence.
- ☐ Related work covers SWE agents, LoRA/QLoRA, trajectory distillation, budgeted agents.
- ☐ Method reproducible (prompts, data, filters, training config, eval).
- ☐ Experiments use paired/held-out design; negative results included.
- ☐ Figures/tables readable; no leaderboard-only claims.
- ☐ Code/data public; licenses compatible; hashes/revisions.
- ☐ 3,000-word limit respected.
- ☐ Submitted before 2026-11-12 23:59 UTC.

Small model

- ☐ E4B/E2B base explicitly declared; single model rule.
- ☐ Prompt/budget tuned for E4B, not copied blindly from 31B.
- ☐ LoRA (if any) validated on the exact base; A/B output differs.
- ☐ Local paired comparison vs E4B baseline.
- ☐ Hidden score obtained.
- ☐ Final-slot opportunity cost considered.
- ☐ Cross-track eligibility understood.